Babel's Machines

How language reshapes the behaviour of multilingual artificial agents

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Roadmap

1. **Why language matters** for the machine-behaviour programme
2. **What LLM agents do** across languages — including the contradictions
3. **Why** — mechanism, alternatives, and the causal loop we still need to close
4. **So what** — evolution, governance, and four falsifiable predictions

LLMs no longer just produce text. They act.

- Negotiate trades and form coalitions (Diplomacy, bargaining)
- Cooperate and defect in iterated games (Prisoner's Dilemma)
- Deceive in social-deduction games (Werewolf, Avalon)
- Manage budgets in auctions; collaborate on hours-long tasks

This is an *ethology*, not a benchmark. The right tools come from biology, evolutionary game theory, and cultural evolution — not engineering.

Rahwan et al., Nature 2019 — the machine-behaviour programme

Our claim, in one sentence

> *"The language an LLM agent operates in systematically reshapes its strategic behaviour, theory-of-mind performance, and prosocial defaults."*

We call this the **Algorithmic Sapir–Whorf Switch** (after Wang et al., 2025).

What we are *not* claiming

Not the strong Whorfian claim that language determines thought.
Just: language-specific representations inside the model modulate the policies the model executes in interactive environments.
Empirical, not metaphysical. We devote real space to evidence against the claim.

Seven families of interactive games — all show language effects

- Social dilemmas (IPD, Public Goods)
- Negotiation (Diplomacy, bargaining)
- Persuasion / auction
- Social deduction (Werewolf, Avalon, Spyfall)
- Cooperative communication (Codenames, Hanabi)
- Open-world games (Minecraft, BALROG)
- Dialogue games (clembench)

Every family has been run multilingual. Every family reports language-dependent divergence.

Same model, same game, different language → different strategy

	ALLC	ALLD	TFT	WSLS
English	26	25	25	24
French	27	18	26	29
Arabic	**32**	**28**	19	21
Chinese	31	30	19	**20**
Vietnamese	26	24	23	27

Aggregated across 4 model families,
≥ 200 rounds/condition.
Descriptive proportions, not meta-analysis.

Three patterns to notice:

- **Arabic is bimodal** — both highest cooperation *and* second-highest defection. Volatile, not adversarial.
- **Vietnamese subverts the collectivist prior** — high defection in some setups; one model collapses to 2% cooperation.
- **Chinese prefers Win-Stay-Lose-Shift** over Tit-for-Tat — outcome-driven, not reciprocity-driven.

Model-family effects also swamp language effects: GPT-4o "hawkish", Mistral "dovish" — regardless of language.

LLMs are "nicer than humans" — in a very specific regime

Frontier LLMs cooperate at rates well above typical human players in iterated Prisoner's Dilemma.

The regime where this holds

Cooperative-payoff matrix ($R{=}3, S{=}0, T{=}5, P{=}1$),
horizon disclosed (e.g. 100 rounds),
opponent is **exploitable** (random, defection prob. 0.5–0.7).

The regime where it inverts

Unknown horizon? Cooperation universalises across all languages.
Pure-competitive game (rock-paper-scissors)? LLMs play *deeper* than humans.

The prosocial default is real but conditional — a property of cooperative & mixed-motive games, not a uniform LLM trait.

Foregrounding contradictions — bounding the claim honestly

Four findings push back against a strong reading of language effects:

1. **Persuasion null** (Biswas et al., CHI 2025): cross-lingual persuasive co-writing — usage shifts, but *no aggregate effect on persuasiveness*. What mattered was beliefs about AI authorship.
2. **Cheap talk dominates** (Madmoun & Lahlou, 2025): one extra communication word raises Stag-Hunt cooperation from **0% to 96.7%** — in every language tested.
3. **Framing dominates** (Buscemi et al., 2025): when game horizon is unknown, cooperation rates universalise across languages.
4. **Vietnamese "defection" = arithmetic error** (Huynh et al., 2025): low-stakes "defection" is sometimes payoff miscomprehension, not strategy.

Language matters, but not as a fixed cultural dial. It interacts with framing, stakes, communication, and tasks.

Two mechanisms account for the cross-lingual gap

Semantic Hub (Wu et al., 2024)

Multilingual inputs project into a centralised, English-dominant intermediate hub.

Causal evidence: English-derived activation steering vectors shift Spanish/Chinese outputs as effectively as native steering. The hub is causally engaged, not vestigial.

Translation Barrier *(Bafna et al., 2025)*

Many failures are late-layer realisation errors — the model reasoned correctly internally, then failed projecting into the target language.

The calibration crisis

Expected Calibration Error

Language	ECE
English	4.6%
Non-English	23.1%

Ahuja et al. 2022 (MEGA / XGLM, BLOOM, PaLM, 15-bin equal-width)

Models are confidently wrong in non-English contexts.

We don't claim a monopoly — three live alternatives

1. **Language-specific microcircuits** (Pires, Schlinger, Garrette 2019 on mBERT)
   Some failures persist for specific language pairs. Suggests partially separated subnetworks.
2. **Subspace interference** (XLM-R, mT5 lineage)
   Scale alleviates but doesn't eliminate cross-lingual interference. Low-resource langs drift.
3. **Latent-language priors** (Wendler et al., 2024)
   Llama decoders pass through an intermediate that is statistically *closer to English* than to the input language.

The natural test — runnable *now*

Language-prioritised models: CT-LLM (800B Chinese) and TigerLLM (Bangla).

If the Semantic Hub is about *training data* and not architecture, these should *invert* the asymmetry.

Closing the loop — from representation to behaviour

The link from "hub asymmetry" to "Vietnamese IPD strategy distribution" is currently **inferential, not causal**.

Why this is fixable now

Recent interpretability work has all the precedent we need:

- Spatial reps causally affect prediction (Chen et al., 2023)
- World-model state vars are linearly decodable *and* causally engaged (Zhang, 2026)
- Layer-resolved tools (LogitLens4LLMs, arithmetic-trajectory probes, syntax/semantics probes)

Next experiment: derive a steering vector from *English* cooperative vs. defective IPD play. Apply to the same model in Arabic. Measure the resulting strategy shift.
This converts a representation claim into a behavioural lever.

Bilingual children get *better* at ToM. LLMs get worse.

Human baseline: bilingual children pass false-belief tasks *earlier* than monolinguals. *(Kovács 2009; Goetz 2003 — the executive demands of code-switching scaffold ToM.)* ***The implicit prediction for machines:*** *multilingual models should be stronger ToM agents, not weaker.*

What we actually see

- ***XToM*** *(Chan et al., 2025): GPT-4o has an 11-point ToM gap between Japanese and Chinese on the same items.*
- ***LLM-Hanabi****: when first-order ToM erodes, both cooperation and deception erode together.*

Caveat (Ullman 2023; Sap et al. 2022): absolute LLM ToM scores are contested. But cross-lingual differences on parallel items are well-defined regardless of construct-validity disputes.

Language is one modulator. There are at least three others.

Personality steering

Ong et al. (2025).
Steering Big-Five Agreeableness modulates IPD cooperation *and* increases exploitation susceptibility.

⇒ *Internal reps can causally steer cooperation.*

Communication delay

Nishimoto et al. (2026).
U-shaped relationship between delay and cooperation. Agents *spontaneously* exploit slower partners.

⇒ *Infrastructure changes behaviour, no language change.*

Partner identity

Jiang et al. (2025).
Human cooperation against an agent depends on its declared identity (human/rule-AI/LLM).

⇒ *Language is one signal that frames identity.*

Any clean test of linguistic relativity must hold these three constant. Otherwise we mistake moderator effects for language effects.

LLM cooperation doesn't fit Axelrod's mould

Axelrod's tradition: cooperation is an *achievement*.
Stable cooperation requires repeated interactions, reputation, punishment, network reciprocity.

LLMs arrive *pre-cooperative*

Before any game is played:

- **Pretraining corpora** = cumulative cultural products of millions of human authors
- **RLHF** = a layer of selected human preferences

These act as **cultural inheritance** (Boyd & Richerson 1985; Henrich 2015).

LLM cooperation is *culturally inherited*, not evolved through interaction. The corollary is testable.

The WEIRD problem and the multilingual transmission bottleneck

WEIRD-skewed corpora

Henrich, Heine, Norenzayan (2010); Muthukrishna et al. (2020):

WEIRD populations are **statistical outliers** on individualism, analytic thinking, fair-procedure framings. If LLMs inherit cooperation from these corpora, they encode WEIRD norms while *presenting as universal*.

Multilingual deployment risks normative imposition — not just performance gaps.

Machines now subject to cultural evolution

LLM populations are linked through:

- Shared benchmarks
- Shared model families
- Distillation & self-training on each other's outputs

If the bottleneck is English (lingua franca of benchmarks + RLHF annotators), **non-English traits drift independently**. Empirical signal: the English–Arabic gap in TCG-Bench at 32B parameters is *not* corrected by scale.

Three societal consequences — already shipping

Unequal institutional quality

Calibration is $\sim 5\times$ worse outside English.

Public-facing agents in the Global South will offer *confidently wrong* advice to those with fewest alternatives.

Already deployed: UlizaLlama (health, Swahili), Aya Expanse (23 langs incl. VI), Bhashini (India).

Uneven attack surface

Code-Switching Red-Teaming (Yoo et al., 2025): **+46.7%** attack success by mixing languages at token level.

Agent-in-the-Middle attacks exploit inter-agent channels.

Adversaries gain leverage where safety data is sparsest.

Co-evolution

LLM outputs re-enter human discourse, then future training corpora.

Language-modulated machine defaults shape the text humans write next.

The cultural-inheritance loop becomes **bidirectional**.

Three governance levers — no breakthroughs needed

1. **Diversify benchmarks.**
   Pair safety benchmarks (M-ALERT) with interactive multi-agent ones (clembench, MultiAgentBench, TCG-Bench). Report behavioural *and* safety gaps *together*.
2. **Multilingual RLHF coverage standards.**
   Every model card reports annotator composition by language and dialect. Suggested floor: $\geq 5\%$ preference labels per language for any model deployed across > 5 languages, with documented dialectal stratification.
3. **Audit standards for multi-agent deployments.**
   Code-switching probes and Agent-in-the-Middle threat models become a *default* test bed — the way prompt-injection probes already are for single-agent products.

What is missing is the *institutional infrastructure*, not the methodology.

Four predictions, each falsifiable with current methods

P1 — Corpus $\leftrightarrow$ cooperation

Cooperative-discourse density in pretraining corpora ρ IPD cooperation rate across languages.
Workarounds for closed corpora: open-data models (Pile, CT-LLM, TigerLLM), proxy corpora, controlled small-model runs.

Falsifier: Spearman ρ indistinguishable from 0 across ≥ 8 languages.

P2 — Hub probe + behavioural readout

Probe layers $\lfloor L/3 \rfloor, \lfloor 2L/3 \rfloor$ to classify hub vs. realisation failures.
Then: apply English-derived steering vector to non-English IPD prompt; measure strategy shift.

Falsifier: no behavioural shift after steering.

P3 — Distillation drift

Controlled distillation chain: traits propagate with high fidelity in English, degrade in lower-resource languages.

Falsifier: no degradation $\Rightarrow$ inheritance-via-distillation is wrong.

P4 — Evaluation reliability ceiling

Multilingual LLM-as-Judge with ≥ 3 heterogeneous judges + self-translation + checklist rubrics raises Fleiss' $\kappa > 0.5$.

Falsifier: saturates below $0.5 \Rightarrow$ need bilingual human adjudication, not better aggregation.

Five things to remember

1. **Language modulates machine behaviour.** It is an active variable, not a passive medium.
2. **The mechanism is architectural.** Hub, Barrier, microcircuits, latent-language — not cultural essentialism.
3. **Cooperation is inherited, not evolved.** Pretraining + RLHF are the inheritance mechanism.
4. **Contradictions are informative.** Persuasion null, cheap-talk dominance, Vietnamese arithmetic errors — they bound the claim.
5. **Synthesis → research agenda.** Four falsifiable predictions; closing the representation–behaviour loop is the most urgent.

Where this sits in the machine-behaviour programme

Continuous with:

Rahwan (machine behaviour) · Han (evolutionary game theory for multi-agent emergence) · Perc (statistical physics of cooperation) · Boyd, Richerson, Henrich (cultural evolution) · Axelrod, Nowak (cooperation without design)

Distinct contribution:

- Treats *language* as a modulating variable in machine ethology
- Reframes "nicer than human" as cultural inheritance, with WEIRD corollaries
- Bridges mechanistic interpretability with population-level cooperation theory
- Supplies an immediately falsifiable agenda

Machine populations now offer a new — uncomfortably consequential — empirical substrate for the science of cooperation.

Thank you

Questions & discussion

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